TWO IDEALS THAT CONTAIN POWERS OF EACH OTHER GIVE THE SAME COMPLETION OF EVERY MODULE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Containment of powers of ideals gives maps between the quotients. Used at the two displayed systems of maps in the counted proof (lines 23035-23037). Produces a map M/I^nM to M/J^{floor(n/c)}M and the symmetric one, from I^c contained in J and J^d contained in I.
- The completion as the inverse limit of the quotient system. Used at the last line of the counted proof (line 23038). Lets two compatible systems of maps between the towers induce mutually inverse maps between the completed modules.

Lemma 0.1. *Let R be a ring. Let I, J be ideals of R. Assume there exist integers $c, d > 0$ such that $I^c \subset J$ and $J^d \subset I$. Then completion with respect to I agrees with completion with respect to J for any R-module. In particular an R-module M is I-adically complete if and only if it is J-adically complete.*

Proof. Consider the system of maps $M/I^nM \to M/J^{\lfloor n/c \rfloor}M$ and the system of maps $M/J^mM \to M/I^{\lfloor m/d \rfloor}M$ to get mutually inverse maps between the completions. □

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